

University of Al-Ameed
College of Medicine
Department of Pharmacology

Pharmacokinetics

(Drug Administration)

BY

Professor
Dr. Haidar Al-Muthaffar

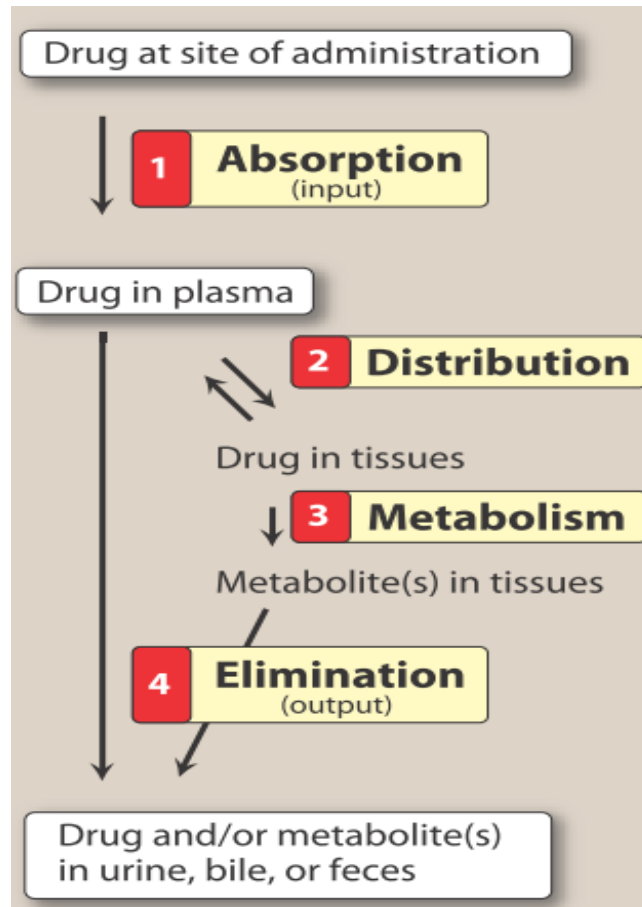
MBChB, MSc, PhD (Clinical Pharmacology)

Objectives:

1. Define pharmacokinetics and explain major divisions
2. Discuss routes of drug administration
3. Describe major characteristics of each route of drug administration

Pharmacokinetics

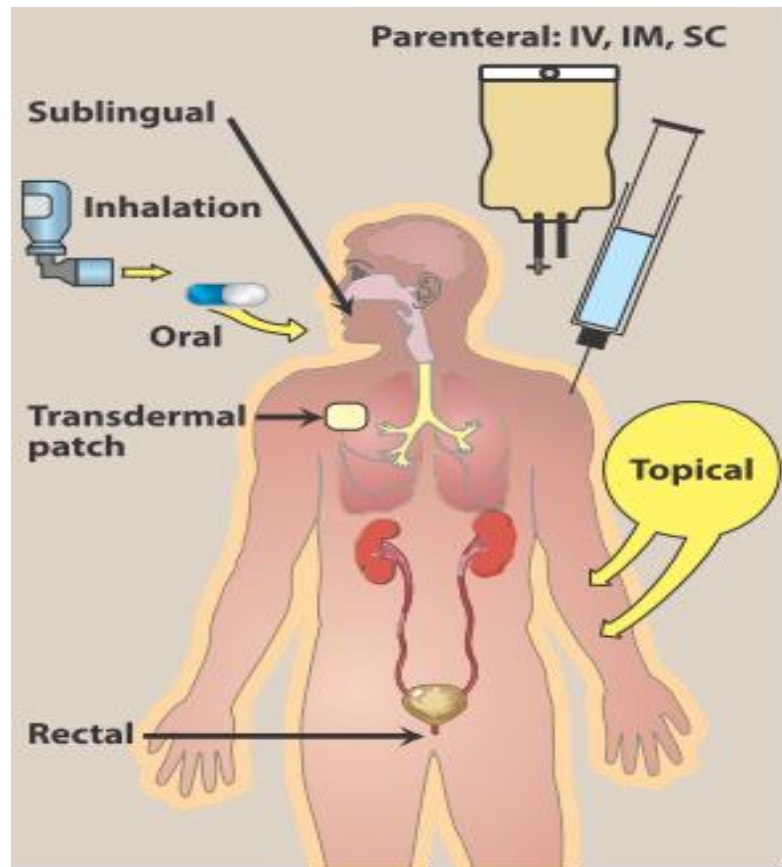
Pharmacokinetics refers to what the body does to a drug. **Four pharmacokinetic properties** determine the **speed of onset of drug action**, the **intensity of the drug's effect**, and the **duration of drug effect**



Administration:

ROUTES OF DRUG ADMINISTRATION:

There are different routes for drug administration. The specific route is determined primarily by the **properties of the drug** (e.g., **water or lipid solubility**, **ionization**) and by the **therapeutic objectives** (e.g., the **desirability of a rapid onset of action**, the **need for long-term treatment**, or **restriction of delivery to a local site**).



A-Enteral route (administering drug to circulation by mouth):

It is the **safest** and **most common**, **convenient**, and **economic** method.

1- Oral route (swallowed drug):

Advantages:

- ✦ Easily self-administered.
- ✦ Low risk of infections (unlike injecting drugs).
- ✦ Toxicities and overdose by the oral route may be overcome with adsorbent antidotes like, activated charcoal. Or by GIT decontamination (e.g. Emesis or Gastric lavage)

Disadvantages:

- ✦ Drug absorption is more complicated.
- ✦ Low pH of the stomach may inactivate some drugs.

Dosage form preparations used could be **liquids** (e.g. **syrup**, **suspension**, **elixir** ..etc.) or **solid forms** (e.g. **tablet**, **capsule**, ..etc.)

Specific preparations:

A-Enteric-coated preparations:

An enteric coating is a **chemical envelope** that **resists the action of gastric juice** but **dissolves readily in the upper intestine**. It used to **protect drugs that are acid unstable** as omeprazole and used to **avoid irritant effect of the drugs** like aspirin.

B-Extended-release preparations:

There is **gradual release** of the drug from the preparation

Advantages:

- Having a **longer duration of action** that **improve patient compliance** (reduce the frequency of drug administration).
- Maintain **drug concentrations** within a **therapeutic range** over a **long period of time**
- They are advantageous for **drugs with short half-lives** like morphine.

2- Sublingual route:

The drug placed **under the tongue** to facilitate direct absorption to systemic circulation.

Advantages:

- **Rapid absorption** (results in *rapid onset of action*).
- **Avoidance of first-pass metabolism** (*increase bioavailability*).
- **Convenience of administration.**
- **Bypass of harsh GI environment.**

B-Parenteral route (administering drug to circulation by injection):

- **Indications:** It is used for **drugs** that are **poorly absorbed from the GI tract** (e.g., heparin) or **unstable in the GI tract** (e.g., insulin) and for **patients** who are **unconscious** or **under circumstances that require a rapid onset of action**.
- **Advantages:** have the **highest bioavailability** and **drugs are not subject to first-pass metabolism** or **harsh GI environment**.
- **Disadvantages:** **pain, fear, local tissue damage, & infections.**

1- Intravenous (IV):

- It permits **rapid effect** and **maximum control** of **circulating levels of the drug** and include:
- ✚ **bolus dose administration** (the **initial full amount** of a drug is **delivered to the systemic circulation almost immediately**) resulting in peak plasma concentration of the drug.
- ✚ **IV infusion** (giving the drug **during a longer time and slower rate**), resulting in a decrease in the peak plasma concentration and an increase in the time the drug is present in the circulation.

Disadvantages:

- **Toxicity** by IV line is **not easily to be overcome** as oral route.
- It **need equipment** and **experienced staff**.
- IV injection may **introduce bacteria or viruses**.
- It may **precipitate blood constituents** or **induce hemolysis**.

2- Intramuscular (IM):

Drugs administered include **aqueous solutions**, (e.g. **diclofenac ampule**) and **powder preparations** that need reconstruction (e.g. **ampicillin vial**) which are **absorbed rapidly** or **specialized depot preparations**, which are **absorbed slowly** providing a sustained dose over an extended period of time (e.g. **methyl prednisolone vials & ampules**).

3- Subcutaneous (SC):

- ✓ This route is **slower than the IV** route
- ✓ SC injection **minimizes the risks of hemolysis or thrombosis** associated with IV injection and may **provide constant, slow, and sustained effects** (e.g. **insulin**).
- ✓ Injecting **Irritant drugs** may produce **severe pain & necrosis**.

C- other routes:

• Inhalation:

- Provide **rapid drug delivery** across the **large surface area** of the **mucous membranes** of the **respiratory tract and pulmonary epithelium**
- Used for **drugs that are gases** (e.g., nitrous oxide general anesthetic) and those that **can be dispersed in an aerosol** (e.g., salbutamol).
- This route is **effective and convenient** for patients with **respiratory diseases** {such as **asthma** or chronic obstructive pulmonary disease (**COPD**)}

• Intrathecal:

Introduce drugs **directly to the cerebrospinal fluid**. E.g., intrathecal amphotericin B is used in treating cryptococcal meningitis (because BBB limits drug absorption)

• Topical:

It is used when a local effect of the drug is desired. For example, **applying a cream directly to the skin**, the use of **nasal drops**, **eye drops**, **scalp lotion or shampoo**, antiseptics, powder....etc.

• Transdermal:

- This route achieves **systemic effects** by **application of drugs to the skin**, usually via a **transdermal patch**.
- The **rate of absorption varies** according to **physical characteristics of the skin at site of application** and **lipid solubility of the drug**.
- This route is used for **sustained delivery of drugs**, such as the antianginal drug nitroglycerin and nicotine transdermal patches, which are used to **facilitate smoking cessation**.

- **Rectal:**

- ✶ Because 50% of the drainage of the rectal region bypasses the portal circulation, the biotransformation of drugs by the liver is minimized with rectal administration.
- ✶ The rectal route has the additional advantage of preventing the destruction of the drug by intestinal enzymes or by low pH in the stomach. The rectal route is also useful if the drug induces vomiting when given orally or if the patient is already vomiting (e.g. paracetamol suppositories).